Aleksandr Kurchev

Senior C++ / Unreal Engine 5 Engineer

Novosibirsk, Russia
Email: a.kurchev4698@gmail.com
Site: https://bezymnijbog.github.io
GitHub: https://github.com/BezymnijBog

Professional Summary

Senior C++ Software Engineer with 6+ years of experience building real-time 3D applications, geometry-heavy software, and Unreal Engine systems. My background combines modern C++, computational geometry, real-time rendering, and software architecture with a strong foundation in physics and mathematics. I enjoy solving technically challenging problems where analytical thinking and domain knowledge matter as much as implementation.

Strong expertise in modern C++, Unreal Engine architecture, performance optimization, and large-scale software development. Experienced in owning complex features from architecture to production, refactoring legacy systems, and collaborating with cross-functional teams.

Passionate about building maintainable, high-performance software and solving technically challenging problems.

Technical Skills

Languages:
C/C++, Modern C++ (up to C++20), Python

Unreal Engine:
UE5 / UE4, Geometry Processing, UMG / Slate, Blueprints, GAS, AI Module, CQTest, PCG, Pixel Streaming

3D & Geometry:
Computational Geometry, OpenGL, CGAL, VTK, OpenCascade

Engineering:
Software Architecture, Performance, Multithreading, Testing, Refactoring

Tools:
CMake, UBT/UAT, Git, SVN, Visual Studio

Professional Experience

C++ / Unreal Engine 5 Software Engineer

TheInvaders
June 2024 - Present

Contributing to a large-scale Unreal Engine 5 platform for Moscow's digital twin.

Highlights

- Worked on a large-scale real-time 3D platform for urban visualization, geospatial interaction, and city environment workflows.
- Developed and maintained core C++ systems for area generation, road and scene logic, geometry processing, and interactive tools.
- Integrated external libraries and data pipelines into Unreal Engine runtime systems to support generation and planning features.
- Improved stability and maintainability through bug fixing, API updates, automated tests, and support for engine and build changes.

Tech Stack: C++20 • Unreal Engine 5 • Python • Git

C++ Software Engineer

LEDAS R&D Services
December 2019 - June 2024

Commercial 3D/CAD software development, with a focus on C++, computational geometry, mesh processing and algorithm development.

Orthodontic Treatment Software Commercial software suite covering the complete clear-aligner workflow, from intraoral scan processing and treatment planning to manufacturability validation and 3D-printable mold generation.

Highlights

- Worked across a suite of clinical 3D applications used to turn intraoral scans into treatment plans, doctor-facing validation tools, and production molds for clear aligners.
- Improved mesh-processing and scan-preparation workflows, including a substantial upgrade to scan cleanup and trimming for raw dental geometry.
- Co-developed a fast trimline algorithm for planning and verification tools, trading a small amount of precision for production-grade performance and better usability.
- Implemented a new production mold variant with ventilation holes to support a new material workflow and prevent deformation during manufacturing.

Tech Stack: C++17 • Boost • CGAL • VTK • Qt5 • CMake • GoogleTest • Python • Git

CAD computation software with Open Cascade

Highlights

- Joined an early-stage R&D project focused on building a geometric solver for industrial CAD workflows.
- Worked from loosely defined input and output requirements, helping shape technical solutions at the start of the project.
- Developed an algorithm for determining triangle visibility from a given viewpoint as part of the project's geometric core.
- The solution was later optimized and reused more broadly within the product, making it one of my most meaningful contributions on the project.

Tech Stack: C++14 • OpenCascade • STL • CMake • SVN

Geological Visualization Platform Desktop application for visualization and analysis of drilling and geological data.

Highlights

- Developed a desktop visualization tool for drilling and tunnel data, combining 3D rendering of tunnel geometry with property-based geological layer visualization.
- Supported multiple data models in the viewer and implemented flexible visual mapping, including palette-driven interpretation of measured layer properties.
- Helped extend the product with a 2D viewer that displayed tunnel-aligned engineering data as curve-based tracks along a parametric tunnel axis.
- Took the application through its later stages in a very small team, working with a part-time QA engineer to bring the product to completion.

Tech Stack: C++11 • Qt5 • OpenGL • STL • SVN

Education

Bachelor's Degree in Physics

Novosibirsk State University
2016 - 2020

Strong foundation in mathematics, mechanics, and analytical problem-solving applied later in computational geometry and 3D software engineering.

Languages

Russian - Native
English - Upper-Intermediate (B2)